

Introduction

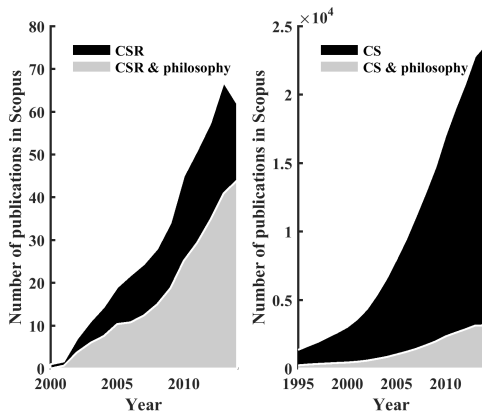
TM the Great Unread

DTL|Digital Arts Initiative
Interacting Minds Centre|Aarhus University

July 25, 2016



Why are we here



“From the dawn of civilization until 2003, humankind generated five exabytes of data. Now we produce five exabytes every two days [...] and the **pace is accelerating**”

Eric Smith

“[I]ncreasingly, scientific breakthroughs will be **powered by advanced computing capabilities** that help researchers manipulate and explore massive datasets.”

Jim Gray

⇒ **4th paradigm** data-intensive science/eScience is and has been a reality for quite a while in the sciences

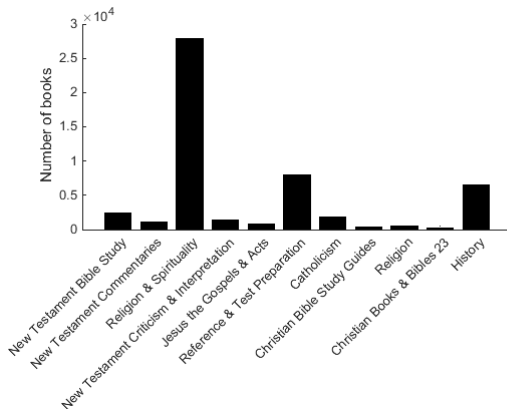
“a limited understanding of the scope, advantages, and limitations of data-intensive methods [...] have created real obstacles to the implementation of this paradigm in historical research”

Text analysis in the humanities

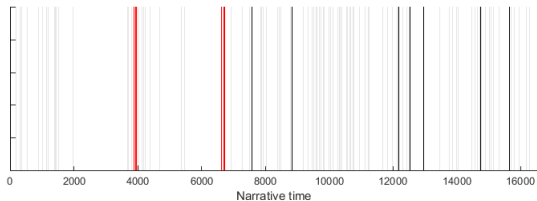
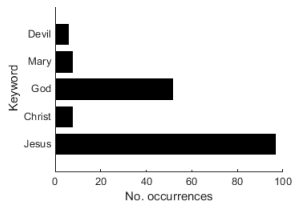
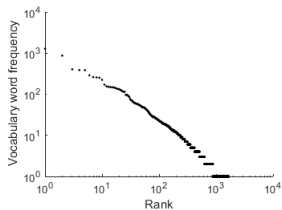


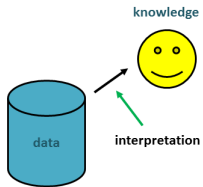
Gospel of Marc (KJV) ~ 16500 words in 16 chp. on 11 p.

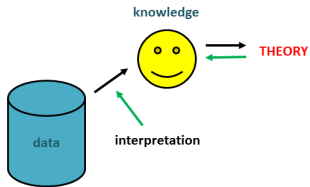
Close reading



Myopic reading



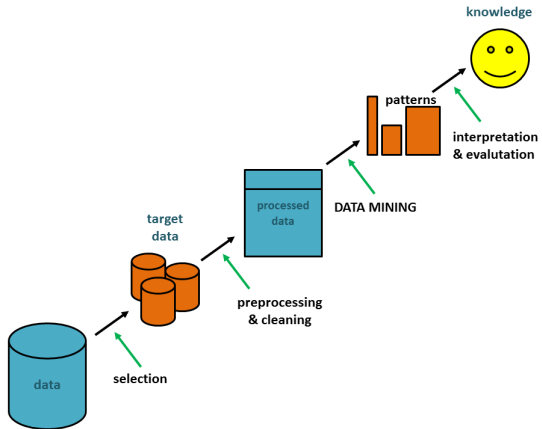


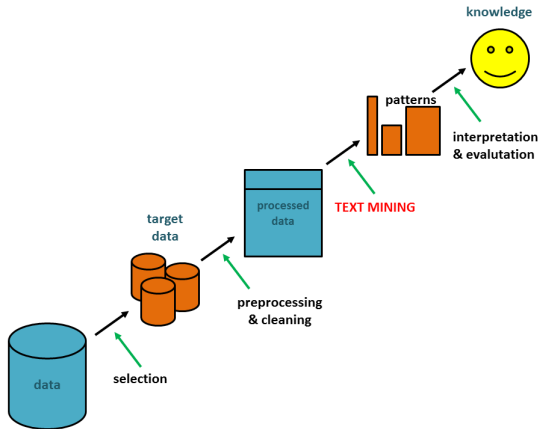


knowledge

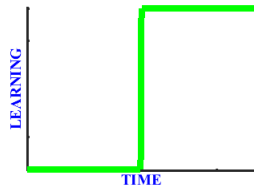
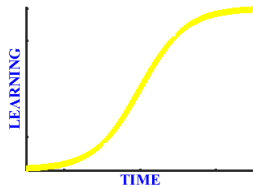
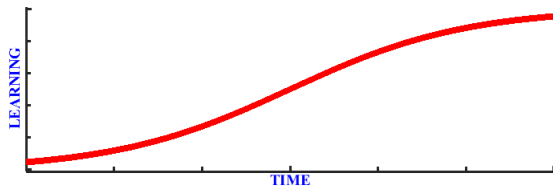


THEORY





Our goal



Introductions



Five minutes one-on-one introductions (who, what, and why)

Communication

BlackBoard (minimum requirement)

Slack (day-to-day course communication)

GitHub (code)

Cloud storage (data and slides)?

Schedule

Time	Monday	Tuesday	Wednesday	Thursday	Friday
9:00-10:30	Introduction	Language models	Design + Communication	Named entities	Machine learning
11:00-12:30	Text analytics	Data preparation	Sentiments	Associations	Clustering
13:30-15:00	R basics	Counting words	Code fest	Code fest	Classification
15:15-17:00	The art of R	In groups	In groups	In groups	Social event

Time	Monday	Tuesday	Wednesday	Thursday	Friday
9:00-10:30		Latent variables	Library*	Unsilos*	Proposal review
11:00-12:30	Social media*	Latent variables	Library*	Temporal dynamics	Proposal review
13:30-15:00	Business analytics*	Network analysis*	Library*	Word embedding*	Proposal review
15:15-17:00		Supervision	Supervision	Supervision	Social event

Week 1 is tutorials and week 2 is research

Ambitious program with buffers

Interdisciplinary challenge

Learning outcomes

- ▶ Demonstrate an ability to delineate and critically evaluate research problems related to text analysis in terms of text mining solutions. This involves accessing previous solution to similar problems
- ▶ Competences in design and implementation of knowledge discovery pipelines that solve research problems related to text analysis. This includes a basic understanding of the various pipeline elements and their dependencies (i.e., data selection, preprocessing and transformation followed by pattern discovery, data mining and evaluation) as well as implementation in open source software
- ▶ Have an understanding of how to communicate projects and findings in accordance with academic and industrial standards

Exams

Not only mixed background, but also mixed course level

Active participation 75% → we have to register you

Alternative to active participation is written take-home assignment

Exam via **web-conferencing is fine**, but we have to know asap

Course material?

Undergraduate

- ▶ 30 minutes
- ▶ 7-point grading scale
- ▶ Prior to the oral examination, students submit a 3-5 page synopsis. **The synopsis may be produced by up to four students.** The grade is awarded on the basis of the oral performance.

Graduate

- ▶ 20 minutes
- ▶ The exam is held as an oral examination on the basis of a synopsis (3-4 pages). **The oral examination can be held in groups of up to 5 persons**

Groups, data and supervision

Project **groups** from tomorrow

- ▶ groups have to define a project and describe it in a synopsis (max 5 persons)
- ▶ both intra- and inter-level are possible
- ▶ fractions are allowed
- ▶ Monday Aug. 1 is abstract deadline

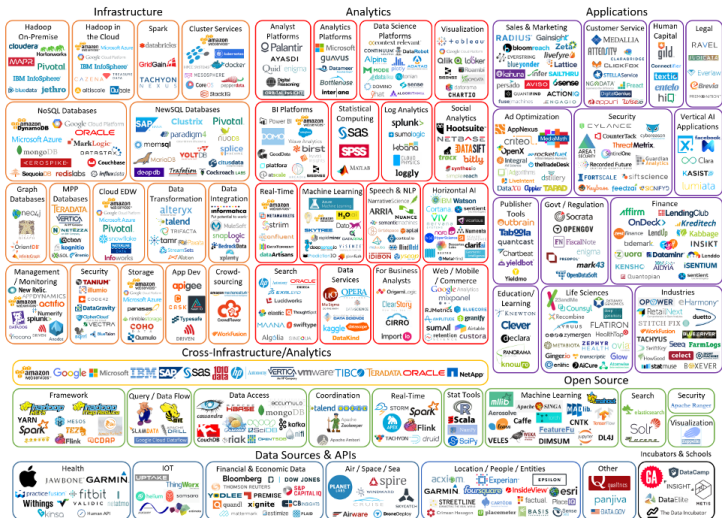
Data set

- ▶ We have a couple of active research data sets
- ▶ Suggest that you make your own

Week 2 **supervision**

- ▶ on the basis of abstract
- ▶ mandatory

data science eco-system





“There is no true interpretation of anything; interpretation is a vehicle in the service of human comprehension. The value of interpretation is in enabling others to fruitfully think about an idea”

Andreas Buja

Social activities

